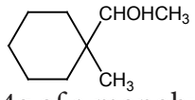
i) $CH_3 - CH_2 - OH \xrightarrow{P\,Br_3} A \xrightarrow{aq.NaOH} B \xrightarrow{Na} C$

ii) $C_6H_5 - OH \xrightarrow{Zn\ dust} A \xrightarrow[Anhydrs\ AlCl_3]{CH_3Cl} B \xrightarrow{acid\ KMnO_4} C$

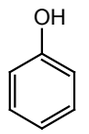
iii) Anisole $\xrightarrow[AlCl_3]{t\text{-butylchloride}} A \xrightarrow{Cl_2/FeCL_3} B \xrightarrow{HBr} C$

iv) 

$\xrightarrow{H^+}$ A $\xrightarrow[ii)\ H_2O]{i)\ O_3}$ B

18. 0.44g of a monohydric alcohol when added to methyl magnesium iodide in ether liberates at STP 112 cm^3 of methane with PCC the same alcohol form a carbonyl compound that answers silver mirror test. Identify the compound.

19. Complete the following reactions

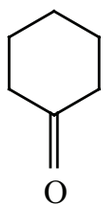
i) 

$\xrightarrow[OH^-]{C_6H_5COCl}$ A $\xrightarrow{Nitration}$ B

(major product)

ii) $C_6H_5\text{-}CHCH(OH)CH(CH_3)_2 \xrightarrow{ConH_2SO_4}$

20. Phenol is distilled with Zn dust followed by Friedel – Crafts alkylation with propyl chloride to give a compound A, A on oxidation gives (B) Identify A and B.

21. $CH_3MgBr +$  $\xrightarrow{H_3O^+}$ A $\xrightarrow{HBr}$ B $\xrightarrow{Mg\ /\ ether}$ C $\xrightarrow{HCHO\ /\ H_3O^+}$ D

Identify A,B,C,D and write the complete equation

22. What will be the product (X and A)for the following reaction

acetylchloride $\xrightarrow[ii)\ H_3O^+]{i)\ CH_3MgBr}$ X $\xrightarrow{acid\ K_2Cr_2O_7}$ A

23. How will you convert acetylene into n-butyl alcohol.

24. Predict the product A,B,X and Y in the following sequence of reaction

butan - 2- ol $\xrightarrow{SOCl_2}$ A $\xrightarrow[ether]{Mg}$ B

butan - 2- ol $\xrightarrow{Cu\ /\ 573K}$ X

B $\xrightarrow{X}$ Y

25. 3,3 – dimethylbutan -2-ol on treatment with conc. H_2SO_4 to give tetramethyl ethylene as a major product. Suggest a suitable mechanism

XII U11-Hydroxy compounds.indd 143 16-12-2022 19:42:26